Linear Algebra II HW 1

1. Prove that there exists a linear transformation $T: R^2 \to R^3$ such that $T(1,1) = (1,0,2)$ and $T(2,3) = (1,-1,4)$. What is $T(8,11)$?

2. Is there a linear transformation $T: R^3 \to R^2$ such that $T(1,0,3) = (1,1)$ and $T(-2,0,-6) = (2,1)$?

3. Determine whether the following are linear transformations from $\mathbb{R}^2$ into $\mathbb{R}^2$:

(a) $T(a_1, a_2) = (1, a_2)$
(b) $T(a_1, a_2) = (a_1, a_1^2)$
(c) $T(a_1, a_2) = (\sin a_1, 0)$
(d) $T(a_1, a_2) = (a_1, a_2, a_1 + 2a_2)$
(e) $T(a_1, a_2) = (a_1 + 1, a_2)$

4. Determine whether the following are linear transformations from P_2 to P_3:

(a) $L(p(x)) = xp(x)$
(b) $L(p(x)) = x^2 + p(x)$
(c) $L(p(x)) = p(x) + xp(x) + x^2 p'(x)$

5. Find the kernel and range of each of the following linear operators on P_3:

(a) $L(p(x)) = xp'(x)$
(b) $L(p(x)) = p(x) - p'(x)$
(c) $L(p(x)) = p(0)x + p(1)$

6. Find the *range* and *kernel* (those are new words for the column space and nullspace) of T.

(a) $T(v_1, v_2) = (v_2, v_1)$. (b) $T(v_1, v_2, v_3) = (v_1, v_2)$.
(c) $T(v_1, v_2) = (0, 0)$. (d) $T(v_1, v_2) = (v_1, v_1)$.

7. Let X and Y be 2-by-2 matrices such that

$$Tr(X)Y + Tr(Y)X = \begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix} \quad \text{and} \quad XY = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}.$$

Find all possible solution for X and Y.

8. Let A be a m-by-n matrix and B be a n-by-m matrix. Show that $Tr(AB) = Tr(BA)$. If $m = n$, is it possible to find 2 m-by-m matrices A and B such that $AB - BA = I_m$.

Solution

1. $(1,0) = 3(1,1) - (2,3), \quad (0,1) = (2,3) - 2(1,1)$

$T(1,0) = 3T(1,1) - T(2,3) = 3(1,0,2) - (1,-1,4) = (2,1,2)$

$T(0,1) = T(2,3) - 2T(1,1) = (1,-1,4) - 2(1,0,2) = (-1,-1,0)$

$T(x,y) = xT(1,0) + yT(0,1) = x(2,1,2) + y(-1,-1,0)$

$(8,11) = 2(1,1) + 3(2,3), \quad T(8,11) = 2T(1,1) + 3T(2,3) = 2(1,0,2) + 3(1,-1,4) = (5,-3,16)$

2. No. Suppose there is such a transformation T. We have

$$-2(1,0,3) = (-2,0,-6)$$
$$-2T(1,0,3) = T(-2,0,-6)$$
$$-2(1,1) = (2,1)$$

Impossible. There is no such transformation T.

3a) No. Because $T(0,0) = (1,0) \neq (0,0)$.

b) No. Suppose there is such a transformation T. We have

$$(4,0) = (2,0) + (2,0)$$
$$T(4,0) = T(2,0) + T(2,0)$$
$$(4,16) = (2,4) + (2,4) = (4,8)$$

Impossible. There is no such transformation T.

c) No. Suppose there is such a transformation T. We have

$$(\pi,0) = (\tfrac{\pi}{2},0) + (\tfrac{\pi}{2},0)$$
$$T(\pi,0) = T((\tfrac{\pi}{2},0) + (\tfrac{\pi}{2},0))$$
$$(0,0) = (1,0) + (1,0) = (2,0)$$

Impossible. There is no such transformation T.

d) $T(ca_1, ca_2) = (ca_1, ca_2, ca_1 + 2ca_2) = c(a_1, a_2, a_1 + 2a_2) = cT(a_1, a_2)$

$$T\big((a_1, a_2) + (b_1, b_2)\big) = T(a_1 + b_1, a_2 + b_2)$$
$$= (a_1 + b_1, a_2 + b_2, a_1 + b_1 + 2a_2 + 2b_2)$$
$$= (a_1, a_2, a_1 + 2a_2) + (b_1, b_2, b_1 + 2b_2)$$

for all $c \in \boldsymbol{R}$, $(a_1, a_2, a_1 + 2a_2), (b_1, b_2, b_1 + 2b_2) \in \boldsymbol{R}^3$

T is linear.

e) No. Because $T(0,0) = (1,0) \neq (0,0)$.

4a) $L(ap(x) + bq(x)) = axp(x) + bxq(x) = aL(p(x)) + bL(q(x))$

for all $a, b \in \boldsymbol{R}$, $p(x), q(x) \in P_2$

L is linear

(b) No. Because $L(0) = x^2 \neq 0$.

c) $L(ap(x) + bq(x)) = ap(x) + bq(x) + axp(x) + bxq(x) + ax^2 p'(x) + bx^2 q'(x)$

$$= a\big(p(x) + xp(x) + x^2 p'(x)\big) + b\big(q(x) + xq(x) + x^2 q'(x)\big)$$
$$= aL(p(x)) + bL(q(x))$$

for all $a, b \in \boldsymbol{R}$, $p(x), q(x) \in P_2$

5a) $xp'(x) = 0 \quad \Leftrightarrow \quad p(x) = a_0, \quad \ker(L) = span\{1\}$

$L(a_2x^2 + a_1x + a_0) = x(2a_2x + a_1), \quad \mathrm{Im}(L) = span\{x, x^2\}$

b) $L(a_2x^2 + a_1x + a_0) = a_2x^2 + a_1x + a_0 - 2a_2x - a_1$

$= a_2x^2 + (a_1 - 2a_2)x + (a_0 - a_1) = 0$

$\Leftrightarrow \quad a_2 = a_1 = a_0 = 0$

$\ker(L) = \{0\}, \quad \mathrm{Im}(L) = span\{1, x, x^2\}$

c) $L(a_2x^2 + a_1x + a_0) = a_0x + a_2 + a_1 + a_0 = 0 \quad \Leftrightarrow \quad a_0 = 0, a_2 + a_1 = 0$

$\ker(L) = span\{x^2 - x\}, \quad \mathrm{Im}(L) = span\{1, x\}$

6a) $T(v_1, v_2) = (v_2, v_1) = (0,0) \quad \Leftrightarrow \quad (v_1, v_2) = (0,0), \quad \ker(T) = \{(0,0)\}$

$\text{Im}(T) = \boldsymbol{R}^2$

b) $T(v_1, v_2, v_3) = (v_1, v_2) = (0,0) \quad \Leftrightarrow \quad (v_1, v_2, v_3) = (0,0, v_3), \quad \ker(T) = \{(0,0, v_3) \mid v_3 \in \boldsymbol{R}\}$

$\text{Im}(T) = \boldsymbol{R}^2$

c) $T(v_1, v_2) = (0,0), \quad \ker(T) = \boldsymbol{R}^2$

$\text{Im}(T) = \{(0,0)\}$

d) $T(v_1, v_2) = (v_1, v_1) = (0,0) \quad \Leftrightarrow \quad v_1 = 0, \quad \ker(T) = \{(0, v_2) \mid v_2 \in \boldsymbol{R}\}$

$\text{Im}(T) = span\{(1,1)\}$

7. Since *Tr* is linear, we have

$$Tr\left[Tr(X)Y + Tr(Y)X\right] = Tr\begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}$$

$$Tr(X)Tr(Y) + Tr(Y)Tr(X) = 0$$

$$2Tr(X)Tr(Y) = 0$$

Either $Tr(X) = 0$ or $Tr(Y) = 0$

Case (i) $Tr(X) = 0$

$$Tr(Y)X = \begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}$$

$$Tr(Y)XY = \begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}Y$$

$$Tr(Y)\begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} = \begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}Y$$

$$Y = \begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}^{-1} Tr(Y)\begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} = Tr(Y)\begin{pmatrix} \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

Let $Y = \begin{pmatrix} 3b & -b \\ -b & b \end{pmatrix} = b\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$

$$Tr(Y) = 4b \neq 0$$

$$X = \frac{1}{4b}\begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix} = \frac{1}{b}\begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}$$

Case (ii) $Tr(Y) = 0$

$$Tr(X)Y = \begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}$$

$$Tr(X)XY = X\begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}$$

$$Tr(X)\begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix} = X\begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}$$

$$X = Tr(X)\begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}\begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix}^{-1} = Tr(X)\begin{pmatrix} \frac{1}{6} & \frac{1}{12} \\ \frac{1}{6} & \frac{5}{6} \end{pmatrix}$$

$$\text{Let } X = \begin{pmatrix} 2a & a \\ 2a & 10a \end{pmatrix} = a\begin{pmatrix} 2 & 1 \\ 2 & 10 \end{pmatrix}$$

$$Tr(X) = 12a \neq 0$$

$$Y = \frac{1}{12a}\begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix} = \frac{1}{3a}\begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}$$

8. Let $A = (a_{ij})_{m\times n}$ and $B = (b_{ji})_{n\times m}$.

$$(AB)_{ii} = \begin{pmatrix} a_{i1} & \cdots & a_{in} \end{pmatrix}\begin{pmatrix} b_{1i} \\ \vdots \\ b_{ni} \end{pmatrix} = \sum_{j=1}^{n} a_{ij}b_{ji}$$

$$(BA)_{jj} = \begin{pmatrix} b_{j1} & \cdots & b_{jm} \end{pmatrix}\begin{pmatrix} a_{1j} \\ \vdots \\ b_{mj} \end{pmatrix} = \sum_{i=1}^{m} b_{ji}a_{ij}$$

$$Tr(AB) = \sum_{i=1}^{m}(AB)_{ii} = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji} = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij} = \sum_{j=1}^{n}(BA)_{jj} = Tr(BA)$$

No. Suppose we can find such matrices A and B such that $AB - BA = I_m$.

$$Tr(AB - BA) = Tr(I_m)$$
$$0 = Tr(AB) - Tr(BA) = m$$

Contradiction!